

Sampling

Reading Objectives:

In this reading, you will learn about sampling and its importance in various applications. You will be introduced to sampling distribution and various distributions associated with it, such as t-distribution and chi-square distribution.

Main Reading Section:

1. Introduction to Sampling

In any statistical understanding related to finding various characteristics related to items belonging to a particular group, which is called “population”

A population containing finite items is known as a **finite population**.

A population containing infinite items is known as an **infinite population**.

A finite subset of the population is called a **sample**.

The number of elements in the population is called **sample size**. If the sample size is greater than or equal to 30, then it is called a “large sample”; otherwise, it is called a “small sample.”

A process of selecting samples (small data) from populations (Big or Huge or Massive data) is called **sampling**. The main purpose of sampling is to analyze populations in an optimum manner and draw valid inferences about them with the help of samples.

Examples:

- . A housewife checks by taking three or four grains of rice (sample) from a cooking pan of whole rice (population) and decides whether the whole rice in the cooking pan is properly cooked or not.
- . A pathologist examines a few drops of blood (sample) taken from the blood of the whole body (population) of a patient and decides the level of illness of the patient.

Parameter and Statistic: The statistical constants of the population are called parameters.

Population mean μ and population standard deviation variance σ are parameters. The statistical constants of a sample are called statistics. Sample mean and sample standard deviation are statistics.

2. Types of Sampling

Random Sampling: Sampling in which the sample items are selected randomly from the population is called random sampling. This is mostly used type in engineering and sciences.

Stratified Sampling: If the population is not homogeneous, then we divide it into various homogeneous strata, and then from each strata, we select the items into the sample randomly. This process of sampling is known as stratified sampling.

Purposive Sampling: In this sampling, items are selected into a sample from the population with a predefined purpose.

3. Sampling Distribution

A sampling distribution is a set of collections of all calculated values of a statistic where each value is obtained from each sample of total possible random samples having size n that can be taken from a population. The sampling distribution follows probability distribution.

If a random sample of size n is taken from a population having the mean μ and known variance σ^2 , then the sample mean $\bar{x}$ is a statistic that serves as an estimator to estimate μ and is a random variable whose sampling distribution has mean $E(\bar{x}) = \mu$.

For samples from infinite populations, the sampling distribution has variance $V(\bar{x}) = \sigma^2/n$.

The standard error is defined as the standard deviation that is the square root of the variance of the sampling distribution of a statistic.

Standard error: $E(\bar{x}) = \frac{\sigma}{\sqrt{n}}$.

Example:

A simple random sample of size 100 is drawn from a population with a standard deviation 20. What is the SE of the sample mean?

Solution

Given $n = 100$ and $\sigma = 20$

Standard Error: $(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{100}} = 2$.

4. Central Limit Theorem

If $\bar{x}$ is the mean of a sample size n taken from a population having the mean μ and the finite variance σ^2 , then $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ is a random variable whose distribution approaches to a standard normal distribution as $n \rightarrow \infty$.

The central limit theorem is better applicable if the sample size $n \geq 30$.

5. t-Distribution

Let $\bar{X}$ be the mean of a random sample of size n drawn from a normal population with mean μ , then $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ is a random variable having t-distribution with parameter $\vartheta = n - 1$ degrees of freedom where s is the standard deviation of the sample.

ν	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$	$\alpha = 0.00833$	$\alpha = 0.00625$	$\alpha = 0.005$	ν
1	3.078	6.314	12.706	31.821	38.204	50.923	63.657	1
2	1.886	2.920	4.303	6.965	7.650	8.860	9.925	2
3	1.638	2.353	3.182	4.541	4.857	5.392	5.841	3
4	1.533	2.132	2.776	3.747	3.961	4.315	4.604	4
5	1.476	2.015	2.571	3.365	3.534	3.810	4.032	5
6	1.440	1.943	2.447	3.143	3.288	3.521	3.707	6
7	1.415	1.895	2.365	2.998	3.128	3.335	3.499	7
8	1.397	1.860	2.306	2.896	3.016	3.206	3.355	8
9	1.383	1.833	2.262	2.821	2.934	3.111	3.250	9
10	1.372	1.812	2.228	2.764	2.870	3.038	3.169	10

6. Chi-Square Distribution

If s^2 is the variance of a random sample of size n taken from a normal population having the variance σ^2 , then $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$ is a random variable having a chi-square distribution with parameter $\vartheta = n - 1$ degrees of freedom.

Upper critical values of chi-square distribution with V degrees of freedom

V	Probability of exceeding the critical value				
	0.10	0.05	0.025	0.01	0.001
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458
7	12.017	14.067	16.013	18.475	24.322
8	13.362	15.507	17.535	20.090	26.125
9	14.684	16.919	19.023	21.666	27.877
10	15.987	18.307	20.483	23.209	29.588
11	17.275	19.675	21.920	24.725	31.264
12	18.549	21.026	23.337	26.217	32.910
13	19.812	22.362	24.736	27.688	34.528
14	21.064	23.685	26.119	29.141	36.123
15	22.307	24.996	27.488	30.578	37.697
16	23.542	26.296	28.845	32.000	39.252
17	24.769	27.587	30.191	33.409	40.790
18	25.989	28.869	31.526	34.805	42.312
19	27.204	30.144	32.852	36.191	43.820
20	28.412	31.410	34.170	37.566	45.315
21	29.615	32.671	35.479	38.932	46.797
22	30.813	33.924	36.781	40.289	48.268
23	32.007	35.172	38.076	41.638	49.728
24	33.196	36.415	39.364	42.980	51.179
25	34.382	37.652	40.646	44.314	52.620
26	35.563	38.885	41.923	45.642	54.052
27	36.741	40.113	43.195	46.963	55.476
28	37.916	41.337	44.461	48.278	56.892
29	39.087	42.557	45.722	49.588	58.301
30	40.256	43.773	46.979	50.892	59.703
31	41.422	44.985	48.232	52.191	61.098
32	42.585	46.194	49.480	53.486	62.487
33	43.745	47.400	50.725	54.776	63.870
34	44.903	48.602	51.966	56.061	65.247
35	46.059	49.802	53.203	57.342	66.619
36	47.212	50.998	54.437	58.619	67.985
37	48.363	52.192	55.668	59.893	69.347
38	49.513	53.384	56.896	61.162	70.703
39	50.660	54.572	58.120	62.428	72.055
40	51.805	55.758	59.342	63.691	73.402
41	52.949	56.942	60.561	64.950	74.745
42	54.090	58.124	61.777	66.206	76.084
43	55.230	59.304	62.990	67.459	77.419
44	56.369	60.481	64.201	68.710	78.750

45	57.505	61.656	65.410	69.957	80.077
46	58.641	62.830	66.617	71.201	81.400
47	59.774	64.001	67.821	72.443	82.720
48	60.907	65.171	69.023	73.683	84.037
49	62.038	66.339	70.222	74.919	85.351
50	63.167	67.505	71.420	76.154	86.661
51	64.295	68.669	72.616	77.386	87.968

Source: <http://furthermathematicst.blogspot.com/2011/08/151-2-distribution.html>↗

7. F-Distribution

If s_1^2 and s_2^2 are the variances of independent random samples of sizes n_1 and n_2 , respectively, taken from two normal populations having the same variance, then $F = s_1^2 / s_2^2$ is a random variable having the F-distribution with $\vartheta_1 = n_1 - 1$ and $\vartheta_2 = n_2 - 1$

Appendix 4a

5 per cent Points of the *F*-distribution

Column represents degrees of freedom (ν_1) for numerator of *F*-test
 Row represents degrees of freedom (ν_2) for denominator of *F*-test

	1	2	3	4	5	6	7	8	9	10	12	24	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	249.1	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.45	19.50
3	10.13	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.638	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.774	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.527	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.841	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.410	3.230
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.115	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	2.900	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.737	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.609	2.405
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.505	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.420	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.349	2.131
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.288	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.235	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.190	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.150	1.917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.114	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.082	1.843
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.054	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.028	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.005	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	1.984	1.733
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.236	2.165	1.964	1.711
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.265	2.220	2.148	1.946	1.691
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204	2.132	1.930	1.672
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190	2.118	1.915	1.654
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177	2.104	1.901	1.638

Source:

<https://image.slidesharecdn.com/fdistributiontable-141214193007-conversion-gate02/95/f-distribution-table-1-638.jpg?cb=1418585>

